

MOHAMMAD AARISH KHAN

Biomedical Engineering Researcher

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CONTACT

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EDUCATION

University at Buffalo
Biomedical Engineering Major
Honors College | Fall 2026 - Present

Williamsville North High School
Class of 2026
AP Scholar of Distinction, 2025

TECHNICAL SKILLS

Programming: Python, Java, Arduino, R
AI/ML: AlphaFold, PyTorch, Hugging Face, Crew AI, Ollama
Comp. Bio: GROMACS, PLIP, Prodigy, HDock, FastENLOC, Biopython, Benchling
Web Dev: Streamlit, Google Colab
Cloud: Oracle Cloud Infrastructure

AWARDS & HONORS

- NY Academy of Sciences: Emerging Student Researcher of the Year 2025
- ACS NERD Outstanding Poster Award 2025
- Kennedy Space Center SmallSat Conference 2025
- Terra Science Fair Medal of High Honors 2025
- College Board AP Scholar of Distinction 2025

CONFERENCES

- Kennedy Space Center / NASA SmallSat 2025
- iFORE 2025; ACS NERD 2025
- Terra Science Fair; IEEE ISEC 2025
- Global Impact Council Conference

WORKSHOPS & CERTS

- MIT Beaver Works: Autonomous Cars
- Biotech in Action - Biogen / Lab Central / MIT
- Oracle Cloud Infrastructure Gen AI Certification
- Eric Pitman Summer Workshop 2025

RESEARCH EXPERIENCE

CURRENT RESEARCH

- Williams Lab - ADHD Genetics Research** *University at Buffalo | Jan 2025 - Present*
- Use FastENLOC to analyze genetic data for an ongoing study of ADHD genetics.
 - Apply computational genetics pipelines and support research interpretation.
- Nushoor - Water Purification Research Collaborator** *Sep 2024 - Present*
- Design a low-cost water purification system for refugee communities in Bangladesh.
 - Tailor system design to affordability, maintenance, and environmental constraints.

PAST RESEARCH

- Eigen Sciences - Cancer Immunotherapy Researcher** *Jan 2024 - Jun 2025*
- Designed a bispecific DuoBody antibody targeting 4-1BB and PD-L1 for NSCLC.
 - Used AlphaFold, GROMACS, HDock, PLIP, Prodigy, and Biopython for analysis.
 - First author, IEEE ISEC 2025. DOI: 10.1109/ISEC64801.2025.11147433
 - Presented at ACS NERD, iFORE, Terra Science Fair, Global Impact Council, and IEEE ISEC.
- DuoDok - Independent Computational Research Tool** *Personal Project*
- Built a web application that automates portions of the bispecific antibody workflow.
 - Authored and presented a research abstract at iFORE 2025 and ACS NERD 2025.
- High-Altitude Balloon Research** *UB Nanotechnology Lab - Summer Internship*
- Designed electronics and code for a five-student high-altitude balloon mission.
 - Studied solar-panel efficiency across altitude; presented at NASA Kennedy Space Center.

- Environmental Genomics** *Eric Pitman Summer Workshop, University at Buffalo*
- Analyzed Scajaquada Creek water samples using whole-genome analysis and R.
 - Presented findings; gained experience in genomics, bioinformatics, and data analysis.

SELECTED PUBLICATION

Bispecific DuoBody Antibody Targeting 4-1BB on Cancer Cells and PD-L1 on T-Cells
for Immunotherapy in Non-Small Cell Lung Cancer (NSCLC). IEEE ISEC 2025.
M. Aarish Khan et al. | DOI: 10.1109/ISEC64801.2025.11147433

LEADERSHIP & EXTRACURRICULARS

- **Computer Club** President (2025-Present); Vice President (2023-25)
- **National Honors Society** Vice President, Class of 2026 (2025-Present)
Organized NHS Running Event and Co-organized Tech Mentorship Program.
- **Technology Student Association** Secretary & NY State Delegate (2024-25)
- **Global Impact Council** Director of Social Media (2024-25); Writing Team (2023-24)
- **Superintendent Inter-High Action Committee** Co-Chair (2023-Present)
- **Shared Decision Making Committee** Class Representative (2024-Present)
- **Science Olympiad (2022-Present)** Science Honors Society (2023-Present)